Thomas Magallon

Portfolio Management Project

I) Introduction to the selected mutual fund

The selected mutual fund for this project is the JPMorgan Large Cap Growth Fund – I Shares, traded under the ticker SEEGX. It is a U.S.-domiciled, open-end mutual fund managed by J.P. Morgan Asset Management.
This fund was founded in 1992 and today is rated 4 stars by Morningstar and bronze medal in 2024. Nowadays, the total fund of assets is up to a total of $126.09bn diversified across 74 assets. Its NAV (Net Asset Value) is currently $92.89.

According to the fund's prospectus and official marketing materials, the investment objective is long-term capital appreciation. Under normal market conditions, at least 80% of the fund's assets are invested in the equity securities of large, well-established companies whose market capitalizations are within the universe of the Russell 1000 Growth Index. The fund is managed as a non-diversified, actively managed portfolio.

Investment Style and Portfolio Profile

SEEGX follows a large-cap growth investment style. It focuses primarily on U.S. large-cap companies with above-average earnings and revenue growth prospects, typically trading at higher valuation multiples than the broad market. The strategy is benchmark-aware but actively managed: the portfolio management team selects stocks they believe have attractive fundamentals, the potential to exceed market expectations, and positive price momentum.
In practice, the portfolio tends to be concentrated in sectors such as information technology, communication services and consumer discretionary, and often includes mega-cap growth names that dominate the U.S. equity market. The fund's Morningstar category is Large Growth, and its historical risk/return profile is consistent with a high-growth, equity-heavy mandate

Benchmark Used for Evaluation

The fund's official performance benchmark, as stated in J.P. Morgan Asset Management materials, is the Russell 1000 Growth Index. This index measures the performance of the growth segment of the U.S. large-cap equity universe and is therefore style-consistent with the fund's investment mandate.
In our project, SEEGX is primarily evaluated against this style benchmark. In addition, for a broader perspective, we also report comparative metrics versus: (i) the Russell 1000 Total Return index, as a representation of the overall U.S. large-cap market, and (ii) the MSCI World Index, as a proxy for the global developed equity market. These additional benchmarks are used in the ratios and CAPM-based analysis (Sharpe ratio, beta, Jensen's alpha, and Treynor ratio).

Data Sources Used in the Analysis

Two main types of data sources are used in this project: market price data and official fund documentation.

- Market price and return data: Daily historical prices and total return for all the stocks are obtained via the *yfinance* Python library, which retrieves data from Yahoo Finance. These data are resampled to a monthly frequency to compute monthly returns, annualized expected returns, and annual volatilities used in the portfolio construction and evaluation.

- Official fund information: The qualitative description of the fund's objective, investment style, benchmark, and key characteristics is taken from the official J.P. Morgan Asset Management product page for the JPMorgan Large Cap Growth Fund – Shares (SEEGX), as well as from the associated fact sheets and commentary documents.

The process

We searched for all the historical data of our 10 assets. To do so, we used python and the library *yfinance.*

```python
import yfinance as yf
from datetime import date

tickers = ["NVDA", "MSFT", "AAPL", "META", "AVGO", "GOOGL", "AMZN", "TSLA", "MA", "ORCL"]

start_date = "2020-09-01"
end_date = date.today()
intervalle = '1mo'

nom_fichier_excel = "donnees_mensuelles_2020_a_aujourdhui.xlsx"

try:
    print("Téléchargement des données mensuelles (2020 à aujourd'hui)...")
    data_mensuelle = yf.download(tickers,
                                 start=start_date,
                                 end=end_date,
                                 interval=intervalle)['Close']
    print("Téléchargement terminé.")

    data_mensuelle.to_excel(nom_fichier_excel)

    print(f"\n✅ Succès ! Le fichier Excel '{nom_fichier_excel}' a été créé.")
    print(f"   Il contient les données mensuelles depuis 2020 et se trouve dans votre dossier de projet.")
    print(f"   Chemin : C:\\Users\\olivi\\PyCharmMiscProject")

except Exception as e:
    print(f"\n❌ Une erreur est survenue : {e}")
```

This script allowed us to access to all the monthly values of our 10 companies' stock prices and created directly an excel file where the values are organised in a table.

The selected time frame analysis

During the period of the first September 2020 until the first September 2025, our group of technologic stocks experienced several phases of strong market movement. The different events that face our group of stock are not considered abnormal or isolated events. These events are a more natural response to the macroeconomic environment and the structural development that the technology sector has faced during the past five years.

In 2020, the high volatility is explained by the bull run which happens in the previous month between march and august. The market has taken back is the benefit of the previous month, which gets a rectification of their price, has more the tech sector had a feeling of overestimated, which increases the number of sales. Also, the presidential election was being prepared, which created uncertainty. From November onward, the trend reversed with Joe Biden's election and the announcement by Pfizer and BioNTech of their vaccine's effectiveness. These events restored investor confidence and encouraged a strong return to the equity markets. Tesla has also been integrated into the S&P 500 Index, which caused massive technical buying by institutional investors.

In 2022, the decline came from inflation and therefore the rise of the interest rate of the Federal Reserve (FED). This provokes the sales of the "overestimated values" like the technology sector. The market of the technology sector starts to recover in 2023-2024, thanks to the development of artificial intelligence and renewed confidence in the technology sector.

These different events illustrate the real economy and sector trends and not temporary anomalies. This is the reason why we choose not to smooth or exclude data. The data set gives an accurate view of the volatility and the risk faced by technology investors and ensures that the portfolio reflects the real market conditions and not artificial steady data.

II) Construction of the Group Portfolio from SEEGX Top Holdings

Period: September 2020 – September 2025

1) Investment Universe and Group Portfolio

A) SEEGX Top 10 Holdings

We start from the 10 largest positions of the JPMorgan SEEGX fund, which are the only assets allowed in our portfolio: AAPL, AMZN, AVGO, GOOGL, MA, META, MSFT, NVDA, ORCL, TSLA.

These are the stocks for which we collected historical monthly prices and built our risk–return statistics over the period September 2020 to September 2025.

B) Final Group Portfolio (Effective PF)

After the full optimization process, the Group portfolio that is labelled "Effective PF" in the Excel file and highlighted in bold black, invests only in a subset of six stocks.

Table 1 – Universe and Group Portfolio Summary

Ticker	Group PF Weigh	Annual Expected Return	Annual Volatility	Included in Group PF
AAPL	3%	15.48%	24.73%	Yes
AMZN	0%	6.08%	29.95%	No
AVGO	48%	47.13%	33.98%	Yes
GOOGL	14%	24.39%	26.85%	Yes
MA	12%	10.14%	23.65%	Yes
META	0%	19.80%	41.22%	No
MSFT	0%	18.31%	21.45%	No
NVDA	4%	51.17%	48.49%	Yes
ORCL	20%	34.04%	34.28%	Yes
TSLA	0%	21.60%	60.55%	No

- Stocks with 0% weight in the final portfolio are AMZN, META, MSFT and TSLA.

- From the 10 SEEGX holdings, during the optimization, we selected a concentrated mix of 6 stocks that together offer the most attractive balance between expected return and risk for the chosen risk profile.

2) Adjustments on the Return and Risk Metrics

To run a robust portfolio selection, we standardized all inputs so they are comparable and on the same time horizon.

C) Data period and frequency

We use monthly prices from 01/09/2020 onward. The window deliberately starts after the initial COVID shock so that the extreme moves of early 2020 do not distort our risk estimates, while still capturing several market phases (post-COVID recovery, 2022 correction and interest rate crisis, 2023-2024,AI rally).

D) From prices to monthly returns

On the "Expected Return" sheet, for each stock and each month, we compute the simple monthly return. These monthly returns form the basis for average monthly return and monthly variance.

Example (approximate figures from the sheet): AAPL average monthly return was 1.29%; AVGO 3.93%; NVDA 4.26%.

E) Annualising expected returns

To work on an annual horizon, we multiply the monthly mean by 12. The resulting annual expected returns and ranking are:

Table 2 – Annual Expected Returns and Ranking

Rank	Ticker	Annual Expected Return	Comment
1	NVDA	51.17%	Very high
2	AVGO	47.13%	Very high
3	ORCL	34.04%	High
4	GOOGL	24.39%	High
5	META	19.80%	Moderate
6	MSFT	18.31%	Moderate
7	AAPL	15.48%	Moderate
8	TSLA	21.60%	Moderate
9	MA	10.14%	Lower

10	AMZN	6.08%	Lower

F) Annualizing risk: variance and standard deviation

From the same monthly returns, we compute monthly variance for each stock, then annual variance, then annual standard deviation (volatility).

Table 3 – Annual Volatility and Return-to-Risk Ratio

Ticker	Annual Volatility	Annual Expected Return	Return / Volatility
AAPL	24.73%	15.48%	0.626
AMZN	29.95%	6.08%	0.203
AVGO	33.98%	47.13%	1.387
GOOGL	26.85%	24.39%	0.908
MA	23.65%	10.14%	0.429
META	41.22%	19.80%	0.480
MSFT	21.45%	18.31%	0.854
NVDA	48.49%	51.17%	1.055
ORCL	34.28%	34.04%	0.993
TSLA	60.55%	21.60%	0.357

- In parallel, the "Variance-Covariance" sheet holds the 10×10 covariance matrix between all stocks, built from the same monthly returns and expressed on an annualised basis. This matrix is what allows diversification to reduce portfolio risk.

- We used the analyze tool pack in excel to make the var-covar matrix

G) Risk-free rate

In the "Variance-Covariance" sheet, we set a risk-free rate of 4% per year (US bond). This is used for creating the efficient frontier.

H) Summary

Before any optimization, all figures of returns, risk and correlations, are converted to consistent annual metrics over the Sep 2020 – Sep 2025 period. This ensures that the portfolio we build is based on comparable, coherent risk-return inputs.

3) Efficient Frontier of SEEGX Top Holdings

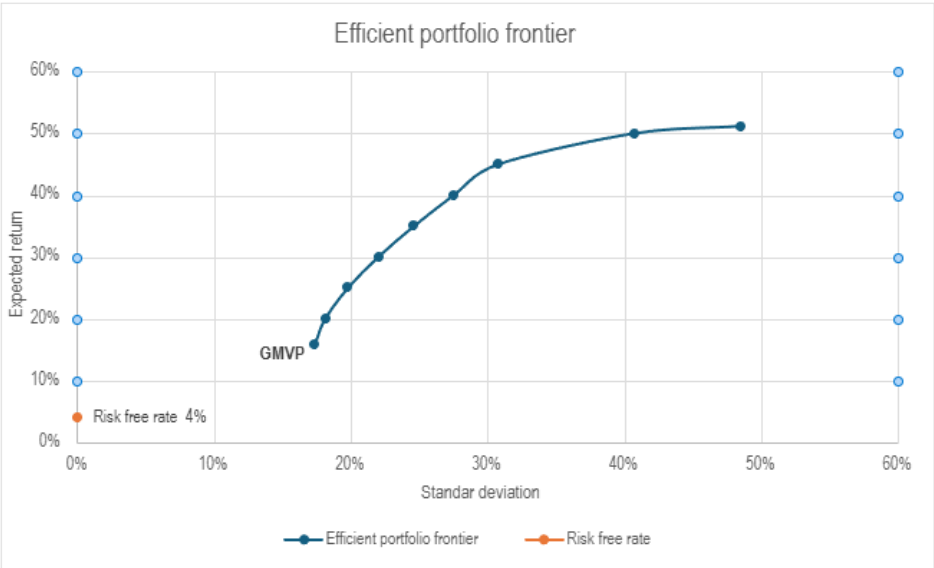
Using the annual expected returns and the covariance matrix, we constructed the efficient frontier under the following constraints: fully invested (sum of weights = 100%) and long only (no short positions).

I) Key points on the frontier

On the "Variance-Covariance" sheet we list several portfolios: the GMVP (Global Minimum Variance Portfolio) and a series of target-return efficient portfolios ("Eff PF, with a specified mean). Each is the minimum-risk portfolio that achieves a given expected return.

Table 4 – Efficient Frontier Summary

Portfolio	Mean Return	Variance	Sigma
GMVP	16%	3%	17%
Eff PF with mean = 20%	20%	3%	18%
Eff PF with mean = 25%	25%	4%	20%
Eff PF with mean = 30%	30%	5%	22%
Eff PF with mean = 35%	35%	6%	25%
Eff PF with mean = 40%	40%	8%	28%
Eff PF with mean = 45%	45%	9%	31%
Eff PF with mean = 51%	51%	12%	35%
Group portfolio (Effective PF)	**36%**	**6%**	**25%**



J) Position of the Group portfolio on the frontier

The Group portfolio (Effective PF) has approximate summary statistics: expected return of 36% and a variance of 6% with a volatility (Sigma) of 25%, and a 4% risk-free rate bond. On the efficient frontier chart, this point lies between the 35% and 40% target-return portfolios: it is a moderate-to-high return portfolio with intermediate risk and a strong risk-adjusted profile.

4) Process for Selecting the Group Portfolio

The efficient frontier gives many best possible portfolios. To choose one portfolio that fits the investor's profile, we followed a two-step decision process.

K) Step 1 – Generate efficient portfolios

Input data: annual expected returns for the 10 stocks, annual covariance matrix, and a risk-free rate of 4%. Constraints: fully invested (weights sum to 1) and no short positions. Optimisation: for each target mean return (20%, 25%, 30%, …, 51%), we minimize portfolio variance to generate the points on the efficient frontier. This step is purely quantitative and creates the menu of efficient options.

L) Step 2 – Use a utility function to select one portfolio

To align the choice with the investor's risk attitude, we introduce a utility score. In the Excel table on the right you have: Column A, a parameter representing the investor's risk aversion (tested at values 200, 400, 600, 700, 750, 800, 1000, 1200) and a Utility column, which increases with expected return and decreases with variance.

For each candidate efficient portfolio, we compute a utility value. Higher utility means the portfolio offers a better trade-off between additional return and additional risk for that level of risk aversion.

Table 5 – Utility Levels and Qualitative Risk Profiles

Risk Aversion (A)	Portfolio Choice	Risk Profile Description	Utility (approx.)
200	Higher-return efficient PF	Very aggressive / high risk	Good but volatile
400	High-return PF	Aggressive	Good
600	Balanced PF on frontier	Balanced – moderate risk	Good
700	**Group portfolio (Effective PF)**	**Balanced growth – aligned with target**	**≈14% (maximum)**

750	Slightly more conservative PF	Cautious growth	Lower
800	More conservative PF	Defensive	Lower
1000	Low-risk PF close to GMVP	Very defensive	Lower
1200	Very low-risk PF	Capital preservation focus	Lowest

When we look at the block labelled "Effective PF", the bold row corresponds to risk-aversion A = 700 and utility » 14% (the highest in that block). For an investor with a risk attitude corresponding to A = 700, the highlighted "Effective PF" is the efficient portfolio that maximises their utility. Any other efficient portfolio either takes too much risk for the extra return, or is too conservative for the return it gives up.

M) Why this particular weight pattern?

The optimization, constrained to the SEEGX universe, leads to strong tilts toward AVGO (48%) and ORCL (20%), which in the historical sample combined high expected returns with attractive diversification properties. It also results in meaningful allocations to GOOGL (14%), MA (12%), AAPL (3%) and NVDA (4%). There is zero weight in AMZN, META, MSFT and TSLA because, given their combination of return, volatility and correlation with the rest, they did not improve the utility score at the chosen risk-aversion level.

Conclusion: We started from the top 10 SEEGX holdings, converted five years of post-COVID monthly data into consistent annual return and risk measures, used these statistics to map all the best risk/return combinations (the efficient frontier), and scored each efficient portfolio using a utility function reflecting our risk tolerance. The portfolio highlighted as "Effective PF" is the one that maximises that utility: it lies on the efficient frontier and offers the most attractive balance of expected return (»36% p.a.) and volatility (»25% p.a.) for the chosen risk profile.

III) The selected mutual fund and the group portfolio evaluation

The purpose of this part is to compare our Group optimal portfolio, the selected mutual fund, the official benchmark and the MSCI World. To evaluate and compare these assets, we will use several ratios such as the Standard deviation, the Sharpe ratio, the beta, the Jensen's alpha and the Treynor ratio. They are all metrics commonly used in finance to compare the performance of different assets, portfolio or funds.
For each metrics used, we will explain how it is built and why it is useful.

- The standard deviation:

This is the most fundamental and direct measuure of total risk. It tells how much the portfolio's returns fluctuated around its average. A high value means a high risk and a low number a predictable low risk path.

The formula is: $\sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \dot{x})^2}{n-1}}$

Where x_i is each monthly return, $\dot{x}$ is the mean of returns and n the number of returns. By comparing the standard deviation to its market risk (beta), we can see how well diversified it is. And the lower the standard deviation is, the safest the portfolio is.

- The sharpe ratio:

This is the most widely used performance measure. It calculates the excess return (portfolio return – the risk-free rate) per unit of total risk (standard deviation)

The formula is: $S_p = (R_p - R_f)/\sigma_p$

Where Rp is the portfolio return, Rf is the risk free rate and σ is the standard deviation of the portfolio's returns.

This ratio is really important and relevant because it makes no assumptions about diversification, it simply measures the final return (premium) against the total risk taken. The higher is the ratio, the better is the portfolio.

- The beta :

The beta is a metric used to measure the systematic risk (market risk) of a portfolio. It indicates the sensitivity of the portfolio's returns to the movements of the market.

The formula is: $\beta_p = \sum_{i=1}^{n} \omega_i \cdot \beta_i$

Where ω_i is the weight of each asset in the portfolio and β_i is the respective beta of each asset.

- Beta = 1: The portfolio moves in line with the market.
- Beta > 1: The portfolio is more volatile than the market ("aggressive" profile).
- Beta < 1: The portfolio is less volatile than the market ("defensive" profile).

Beta is less a performance measure and more a "style" of the risk profile.

- The Jensen's alpha:

The alpha is based on the CAPM. It measures the outperformance/underperformance of a portfolio relative to what was expected, given its level of market risk (beta). It is the return generated by a manager. Or we can say that it evaluates the added value of active management.

The formula is: $\alpha_p = R_p - [R_f + \beta_p \cdot (R_m - R_f)]$

Where Rm is the benchmark return (here the MSCI World index).

- Alpha > 0: The manager "beats the market"; they generated a return superior to what was justified by the market risk they took.
- Alpha < 0: The manager "destroyed value" compared to a simple passive strategy with the same Beta.

The higher is the alpha, the better the added value from active management is.

- The Treynor ratio:

This ratio is close to the sharpe ratio. It measures the excess return per unit of systematic risk. The formula is: $T_p = (R_p - R_f)/\beta_p$

The difference with the sharpe ratio is that the Treynor ratio assumes that the portfolio is already well-diversified and takes in account market risk and not idiosyncratic risk. In our case, this ratio will be the last one to be considered for our portfolio because we only have 10 assets in total and 6 in our efficient portfolio and idiosyncratic risk can not be erased in this case.

	GMVP	Effective PF	Mutual fund SEEGX	Russell 1000	MSCI World index
Sharpe ratio	0,68	1,27	1,45	0,88	1,05
Beta	1,028	1,237	0,98	1,18	1
Jensen's alpha	2,09%	18,51%	3,02%	-1,33%	0,17%
Treynor ratio	0,116	0,260	10,96	12,58	11,92
Standard deviation	17%	25%	15,93%	18,83%	11,46%

Our analysis of the computed metrics reveals a sharp contrast between our two constructed portfolios and the benchmarks, highlighting a classic trade-off between aggressive stock selection and diversification.

The "Effective Portfolio" (PF), constructed with only six stocks, was designed to be an aggressive strategy aiming for high returns. The metrics confirm this aggressive posture but also reveal that while it was exceptionally successful in stock selection, it suffers from a critical lack of diversification.

The portfolio's aggressive strategy is immediately evident from its Beta of 1.237, the highest in the group, indicating it is designed to be significantly more volatile than the market (MSCI World). This risk profile is confirmed by its Standard Deviation of 25%, which is by far the highest volatility of any asset compared. This high number is a direct consequence of both its high market risk (Beta) and, more importantly, a high degree of specific risk due to its 6-stock concentration. It is 9% more volatile than the mutual fund and 14.5% more volatile than the MSCI World.

Where the portfolio truly stands out is its Jensen's Alpha, which at 18.51% is exceptional. This metric measures the "skill" of the strategy, or its ability to generate returns in excess of what is justified by its market risk. This result far outpaces the mutual fund's respectable 3.02% Alpha and proves our stock selection process created substantial value.

However, this raw skill did not translate into the best overall efficiency. The portfolio's Sharpe Ratio of 1.27, while strong enough to outperform the Russell 1000 and MSCI World, fails to beat the Mutual Fund's 1.45. The mutual fund achieved a superior risk-adjusted return precisely because its broad diversification (15.93% volatility) provided a much smoother ride for the returns it generated.

The portfolio's core structural problem is best diagnosed by its Treynor Ratio. At 0.260, this number is abnormally low—approximately 50 times lower than the benchmarks. This metric,

which measures return per unit of market risk, indicates a severe inefficiency. When combined with the high Alpha, it proves that our portfolio's returns were driven almost entirely by specific, non-market risk (our individual stock bets) rather than by an efficient capture of market risk.

In conclusion, our Effective Portfolio is a high-octane, concentrated bet.

Its primary success is its exceptional stock-picking skill, proven by its massive 18.51% Jensen's Alpha. Its primary failure, however, is a critical lack of diversification, which is visible in its high 25% Standard Deviation and its catastrophically low Treynor Ratio.

This "boom-or-bust" profile means the portfolio is suitable only for a high-risk-tolerance investor who is willing to endure extreme volatility in exchange for the potential of high, alpha-driven returns. For the average investor, the SEEGX Mutual Fund proved to be the superior investment, offering a better-balanced and more efficient (higher Sharpe) path to returns.

Conclusion

To summarize, this project successfully constructed a selective investment portfolio derived from the core holdings of the SEEGX mutual fund. The final composition, restricted to six securities, demonstrated an exceptional facility for security selection, evidenced by a colossal excess return (Jensen's Alpha) of 18.51%, which substantially outperformed the benchmark fund (3.02%).

However, this "high-stakes" strategy exhibits a fundamental flaw in asset diversity. The portfolio's sharp concentration led to extreme price variability (Standard Deviation of 25%) and a severely low return per unit of systematic risk (Treynor Ratio). This confirms that the gains originated from specific, non-market related risks rather than an effective capture of broad market movement.

Consequently, while the selection prowess of our portfolio is clear, its overall efficacy (Sharpe of 1.27) proved inferior to the better-balanced SEEGX Fund (Sharpe of 1.45). Our construct is therefore a "highly speculative" alternative, best suited for investors with an extremely elevated risk appetite. Conversely, the mutual fund represents a more prudent and risk-adjusted investment solution for the typical market participant.